

AEKANK PATEL

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Education

Stevens Institute of Technology

Master of Science in Data Science, **GPA: 3.97/4.0**

Hoboken, NJ

Sep 2024 – May 2026

Manipal Institute of Technology

Bachelor of Technology in Mechatronics Engineering, **GPA: 7.83/10**

Manipal, India

Oct 2020 – Jul 2024

Skills

Programming: Python, R, SQL, C

Machine Learning: Supervised Learning, Unsupervised Learning, Model Evaluation, Hyperparameter Tuning, Ensemble Learning

Deep Learning & AI: CNN, RNN, Transfer Learning, NLP, LLMs, Computer Vision, RAG, Explainable AI

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, OpenCV, SciPy, Matplotlib, Seaborn, HuggingFace

Data & Backend: Flask, REST APIs, SQLite, PostgreSQL, MongoDB

Big Data: Apache Spark, Hadoop, YARN, Google Cloud Platform, Google Dataproc, Google Cloud Storage, Parquet, ETL

Tools: Streamlit, Git, Tableau, Power BI, MATLAB

Experience

Stevens Institute of Technology

Graduate Teaching Assistant

Hoboken, NJ

Sep 2025 – May 2026

- Evaluated quantitative and Python-based assignments for 89 graduate students across Numerical Linear Algebra for Big Data, Intro to Probability Theory, and Foundational Mathematics for Data Science, delivering consistent and timely feedback.
- Standardized grading criteria by identifying recurring analytical errors and improving consistency in evaluation, and feedback.

Matrix ComSec Pvt. Ltd.

Research and Development Intern

Vadodara, India

Jan 2024 – May 2024

- Developed a real-time fall detection system using deep learning and computer vision, implementing a hybrid CNN and MediaPipe pipeline with YOLOv5 for multi-person detection and live video monitoring.
- Evaluated CNN, RNN, BodyPix, and R-CNN architectures through literature review and experimentation, then optimized the selected pipeline using TensorFlow, PyTorch, OpenCV, and NumPy for robust real-time inference.
- Annotated 100,000+ images to improve model generalization and achieved 91.39% test accuracy, validating performance on real-world video feeds across varied body orientations and environments.

Nimit Electronics and Equipments

Data Science Intern

Vadodara, India

May 2023 – Dec 2023

- Queried and analyzed 80,000+ surveillance events, access-control logs, and IoT sensor records using SQL, identifying operational trends across incidents, device health, and system activity.
- Developed Python ETL pipelines using Pandas and NumPy to extract, clean, transform, validate, and standardize data from 12 heterogeneous sources for downstream analytics and machine learning.
- Supported incident-classification and anomaly-detection models using Scikit-learn, evaluated performance with accuracy, precision, recall, and F1-score, and created dashboards tracking incident trends, device uptime, and system utilization.

Projects

E-Commerce Product Search and Recommendation System

Oct 2025 – Dec 2025

- Built an end-to-end e-commerce product search and recommendation pipeline on Google Cloud Platform using Apache Spark, Dataproc, and Google Cloud Storage to process 9.8M MARQO-GS-10M query-product records.
- Implemented TF-IDF, LSH-based semantic retrieval, and FP-Growth market basket analysis in Spark MLlib, achieving Precision@K of 0.782, NDCG of 0.856, MRR of 0.790, and MAP of 0.748.

FinRAG: Financial Document Intelligence System [\[Live Demo\]](#) [\[GitHub\]](#)

Jun 2025 – Aug 2025

- Constructed a RAG pipeline over 25+ financial documents using HuggingFace embeddings, vector similarity search, and Groq LLaMA for finance document question answering.
- Obtained 80% retrieval accuracy through metadata filtering, prompt optimization, and confidence scoring; launched the system on Streamlit Cloud.

FRAUDGEN: Unmasking Fraud with Real-Time Explanations [\[Live Demo\]](#) [\[GitHub\]](#)

Mar 2025 – May 2025

- Achieved 96.55% recall and 0.9995 AUC on the PaySim dataset (6.36M transactions, 0.1291% fraud) using XGBoost with engineered behavioral, transactional, and risk-based features.
- Validated model performance using 5-fold stratified temporal splits to address class imbalance and temporal leakage; implemented a real-time system integrating IP geolocation and VPN detection.

Certifications

[Google Data Analytics](#), [IBM AI Engineering](#), [AWS Cloud Foundations](#), [AWS Data Engineering](#), [Bloomberg Market Concepts](#)